Name 1:_____
Name 2:_____

Example

Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

We will prove that f is continuous nowhere. Let $a \in \mathbb{R}$ and suppose $\lim_{x\to a} f(x) = L \neq 0$. Let $\varepsilon = \frac{|L|}{2} > 0$. Given any $\delta > 0$, the interval $(a-\delta, a+\delta)$ contains an irrational number x. For such an x, $|f(x) - L| = |0 - L| = |L| > \frac{|L|}{2} = \varepsilon$, which contradicts the assumption that $\lim_{x\to a} f(x) = L$. This shows that, if $\lim_{x\to a} f(x)$ exists, then it must be zero. Suppose this is the case and take $\varepsilon = \frac{1}{2}$. Given $\varepsilon > 0$, the interval $(a-\delta, a+\delta)$ contains a rational number x. For such an x, $|f(x)| = 1 > \frac{1}{2} = \varepsilon$, which contradicts the assumption that $\lim_{x\to a} f(x) = 0$. This proves that $\lim_{x\to a} f(x)$ does not exist for any $a \in \mathbb{R}$, and therefore f is continuous nowhere.

Name 1:_____

Name 2:_____

Problem of the week

Consider the function

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

At which points is $f(x)$, if any, is continuous? Write your answer as a proposition and prove it.

Proposition. The function $f(x)$ is continuous _____.